

# Exploring Neuromorphic Electronics

## VO2 Memristors and Oscillatory Circuits

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# Part 1: The Challenge and Motivation

## Energy Crisis in Computation

AI training and inference consume megawatts of power due to the **von Neumann Bottleneck**. Moving data between CPU and memory costs  $> 100\times$  more energy than the computation itself.

## Neuromorphic Solution

The brain ( $\approx 20W$ ) achieves efficiency by collocating memory and processing. Memristor crossbar arrays (right) replicate this, enabling **hardware-level single-step matrix-vector multiplication** [3]. This analog approach performs complex operations instantly via physics (Ohm's/Kirchhoff's laws).

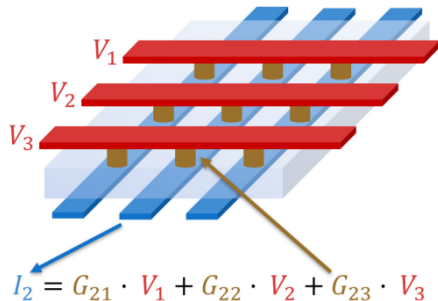


Figure 1: Memristor Crossbar Array

- **Theory:** Vanadium Dioxide (VO<sub>2</sub>) Physics
- **Methods:** Measurement Setup and Objectives
- **Results:**
  - ▶ Voltage-Induced Switching (I-V)
  - ▶ Thermal Phase Transition (R-T)
  - ▶ Relaxation Dynamics (Oscillatory Behavior)
- **Conclusion:** Impact and Future Outlook

# Theory: Vanadium Dioxide (VO<sub>2</sub>)

## Insulator-Metal Transition (IMT)

VO<sub>2</sub> is the premier material for artificial neurons. At room temperature, it acts as an insulator. However, above  $\approx 68^\circ\text{C}$  (or when triggered by Joule heating), it physically transforms into a highly conductive metal.

## Hysteresis and Memory

Crucially, the return path to the insulating state differs from the heating path. This **thermal hysteresis** provides the necessary "memory" effect, characterizing the device as a **Volatile Memristor**. Unlike non-volatile crossbars (synapses), these devices naturally generate spikes, mimicking biological neuron firing [4].

## Switching Ratio

$$R_{ins}/R_{met} > 1000$$

*Volatile Memristor*



# Measurement Session Roadmap

Our characterization session focused on four specific experimental tasks:

## 1. Circuit Validation

Verification of the measurement setup and safety limits using series resistor topology to prevent device burnout.

## 2. $I(V)$ Curve Characterization

Voltage sweeping to identify non-linear switching thresholds (Set/Reset voltages) and current-voltage hysteresis.

## 3. Temperature-Dependent Resistance

Direct observation of the physical phase transition by controlling the sample temperature externally.

## 4. Relaxation Dynamics

Time-domain analysis of the switching speed to evaluate feasibility for oscillatory circuits

# Experimental Setup

## Circuit Topology

We utilized a series circuit configuration:

$$V_{source} \rightarrow R_{series} \rightarrow VO2.$$

The protection resistor  $R_{series} \approx 2.17k\Omega$  is critical. It prevents thermal runaway and destruction when the device switches to its low-resistance state.

## Instrumentation

Current-voltage generation and sensing were handled by an **NI myDAQ** unit. A custom C# application was used to automate the parameter sweeps and data logging.

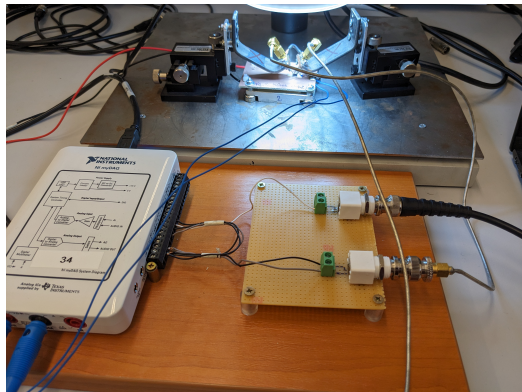


Figure 2: Measurement Setup with Probes

# Result: Voltage-Induced Switching (Task 2)

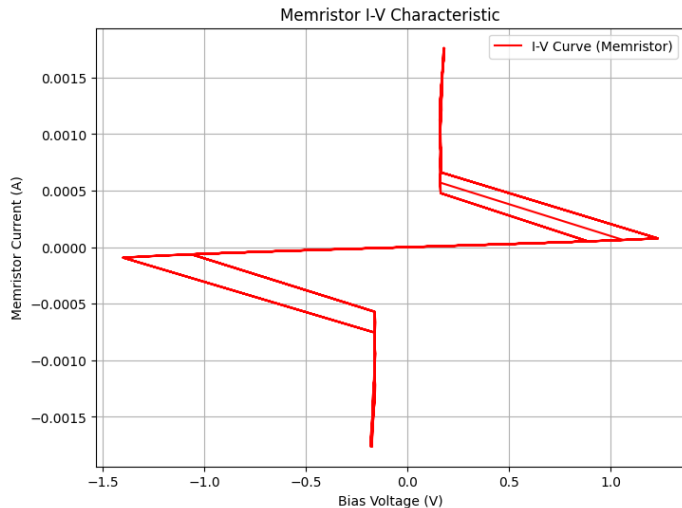
## Threshold Switching

The experimental data reveals a clear hysteresis loop:

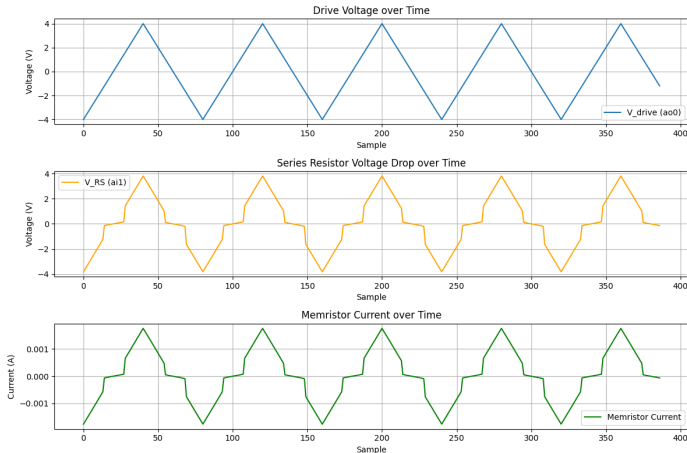
**Set Transition:** At  $\approx 2.5V$ , Joule heating triggers the IMT, causing a sharp current jump.

**Reset Transition:** As bias drops, the device cools and recovers high resistance at a lower voltage.

**Volatility:** The device always returns to the Off state at 0V, confirming volatile behavior.



# Result: Time-Dependent Behavior



**Figure 4:** Time-dependent measurements during the  $I(V)$  sweep: (top) drive voltage, (middle) series resistor voltage drop, (bottom) calculated memristor current.

# Result: Thermal Phase Transition (Task 3)

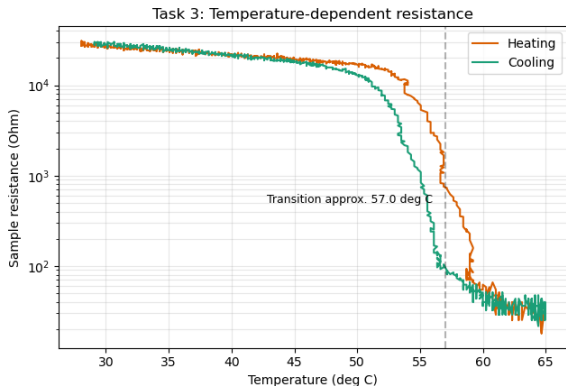


Figure 5: Resistance vs. Temperature

## Physical Verification

To verify the switching mechanism, we heated the sample externally:

**Sharp Transition:** Near  $57^{\circ}\text{C}$ , resistance drops by orders of magnitude.

This confirms that the switching observed in the I-V curves is indeed driven by the Insulator-Metal Transition phase change, rather than purely electronic breakdown effects.

# Result: Activation Energy Analysis

## Arrhenius Analysis

By plotting  $\ln(R)$  against  $1/T$ , we analyzed the conduction mechanism in the insulating state.

## Semiconductor Behavior

The linear region indicates standard thermally activated transport ( $R \propto e^{E_a/k_B T}$ ).

The results are consistent with standard semiconductor physics for VO<sub>2</sub> (typically  $E_a \approx 0.3 - 0.5\text{eV}$ ), confirming the semiconducting nature of the Off state.

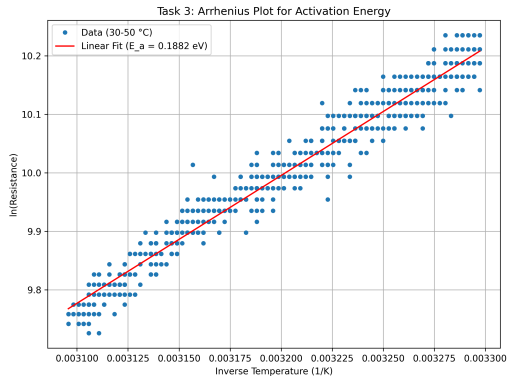


Figure 6: Arrhenius Fit of Resistance Data

# Result: Relaxation Dynamics (Task 4)

## Oscillator Attempt

We aimed to construct a Pearson-Anson relaxation oscillator (Neuristor) by biasing the device in its negative differential resistance region using a capacitor.

## Observations vs. Challenges

We successfully triggered a single switching event, observing the characteristic current spike (shown right).

However, sustaining a stable oscillation train proved difficult. The device exhibited instability and degradation after repeated high-speed cycling. This highlights a key challenge in the field: ensuring device endurance for continuous neuromorphic operation.

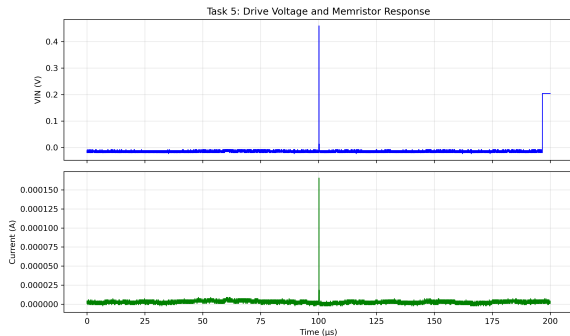


Figure 7: Transient Response to Pulse

# From Lab to State of the Art: Ising Machines

## Connecting Our Measurements to Frontier Research

Our single-device oscillator (Right: Task 4 result) represents the fundamental building block. While maintaining stability for one device was challenging, recent breakthroughs have scaled this concept massively.

### The VO2 Ising Machine

Maher et al. (2024) recently demonstrated a system of **coupled VO2 oscillators** integrated on a CMOS platform [5].

- By coupling many oscillators (like the one we measured), they created an **Ising Machine Solver**.
- This analog computer naturally solves NP-hard optimization problems (e.g., Graph Coloring) by letting the coupled oscillators settle into a minimum energy state (synchronization).



## Stochasticity and Neural Dynamics

In Task 3, we confirmed the thermal origin of the transition. Interestingly, the intrinsic noise and variability we observed are not just "errors"—they are features.

## Stochastic Neurons

Yi et al. (2018) showed that the intrinsic stochasticity in VO2 memristors closely mimics the noise found in biological ion channels [6].

- This "noise" allows artificial neural networks to escape local minima during learning.
- Our verification of the switching physics confirms that these devices are not just digital switches, but biologically plausible, dynamical systems suitable for next-gen probabilistic computing.

# Conclusion

## Summary

We successfully characterized the VO<sub>2</sub> memristor, confirming its potential as a physical building block for neuromorphic hardware.

## Key Findings:

- **Robust Switching:** Demonstrated a volatile On/Off ratio  $> 1000$ .
- **Physical Verification:** Confirmed the thermal origin (IMT) and activation energy.

## Outlook

While the switching physics is sound, improving device durability is the critical next step. Solving this will enable large-scale *Oscillatory Neural Networks* capable of solving complex NP-hard problems using coupled physics [5].

## **Special Thanks to:**

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**Lab Partner**

*Gregorio Jaca*

# References

- [1] Zsigmond Pollner, András Halbritter, Tímea Török et al., *Neuromorphic Electronics Lab Description*, Budapest University of Technology and Economics, 2025.
- [2] Y. Huang et al., “Memristor-based hardware accelerators for artificial intelligence”, *Nature Reviews Electrical Engineering*, 1, 286 (2024).
- [3] F. Aguirre et al., “Hardware implementation of memristor-based artificial neural networks”, *Nature Communications*, 15, 1974 (2024).
- [4] H. Liu et al., “Artificial Neuronal Devices Based on Emerging Materials”, *Advanced Materials*, 35, 2205047 (2023).
- [5] O. Maher et al., “A CMOS-compatible oscillation-based VO<sub>2</sub> Ising machine solver”, *Nature Communications*, 15, 3334 (2024).
- [6] W. Yi et al., “Biological plausibility and stochasticity in scalable VO<sub>2</sub> active memristor neurons”, *Nature Communications*, 9, 4661 (2018).